SATGURU STUDY CENTRE

Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class: 10

UNIT TEST - 2, AUGUST - 2024

MATHEMATICS

Time Allowed : 1.30 Hours] [Max. Marks : 50

PART - I

I. Choose the correct Answer. 7x1=7

1. If 6 times of 6^{th} term of an A.P is equal to 7 times the 7^{th} term then the 13^{th} term of the A.P is

(a) 0 (b) 6 (c) 7 (d) 13

2. The next term of the sequence $3/16, 1/8, 1/12, 1/18, \ldots\ldots$ is

(a) $\frac{1}{24}$ (b) $\frac{1}{27}$ (c) $\frac{2}{3}$ (d) $\frac{1}{81}$

3. If (x-6) is the HCF of $x^2 - 2x - 24$ and $x^2 - kx - 6$ then the value of K is

(a) 3 (b) 5 (c) 6 (d) 8

4. $y^2 + 1/y^2$ is not equal to

(a) $\frac{y^4+1}{y^2}$ (b) $(y+ 1/y)^2$ (c) $(y - 1/y)^2 + 2$ (d) $(y+1/y)^2 - 2$

5. Which of the following should be added to make $x^4 + 64$ a perfect square

(a) $4x^2$ (b) $16x^2$ (c) $8x^2$ (d) $-8x^2$

6. If in ΔABC, DE || BC AB = 3.6 cm, AC = 2.4 cm and AD = 2.1 cm Then the length of AE is

(a) 1.4 cm (b) 1.8 cm (c) 1.2 cm (d) 1.05 cm

7. If (5,7), (3,P) and (6,6) are collinear, then the value of P is

(a) 3 (b) 6 (c) 9 (d) 12

PART - II

II. Answer any five questions only. [Q.No. 14 is compulsory]. 5x2=10

8. In a G.P 729, 243, 81, ... Find t_7

9. Find the First term and Common difference of the Arithmetic Progression whose n^{th} term is $t_n = -3 + 2n$

10. Find the LCM of $5x - 10, 5x^2 - 20$

11. Simplify : $\frac{5t^3}{4t - 8} \times \frac{6t - 12}{10t}$

12. Show that the points P (-1.5, 3), Q (6, -2) R (-3, 4) are Collinear.

13. The line through the points (-2, a) and (9,3) has slope -½ Find the value of a.

14. In ΔABC, D and E are points on the sides AB and AC respectively such that DE || BC. If $\frac{AD}{DB} = \frac{3}{4}$ and AC = 15 cm Find AE.

PART - III

III. Answer any 5. Q.No. 21 is compulsory. **5x5=25**

15. Find the sum of all Natural numbers between 300 and 600 Which are divisible by 7.

16. Find the sum to n terms of the series 3 + 33 + 333 + ...

17. Simplify : $\frac{b^2 + 3b - 28}{b^2 + 4b + 4} \div \frac{b^2 - 49}{b^2 - 5b - 14}$

18. Find the square root of $37x^2 - 28x^3 + 4x^4 + 42x + 9$

19. State and Prove Basic Proportionality Theorem.

20. Find the area of the Quadrilateral formed by the points (-9, -2), (-8, -4), (2, 2), and (1, -3)

21. Find the G.C.D of $6x^3 - 30x^2 + 60x - 48$ and $3x^2 - 12x^2 + 21x - 18$

PART - IV

IV. Answer Any One of the following. **1x8=8**

22. Draw a triangle ABC of base BC = 8 cm, $\angle A = 60^0$ and the bisector of $\angle A$ meets BC at D Such that BD = 6 cm.

23. Construct a ΔPQR which the base PQ = 4.5 cm, $\angle R = 35°$ and the median RG from R to PQ is 6 cm.